

Causal Fairness Analysis

Purpose

The purpose is to understand “why” a fairness issue exists, to identify the Root causes of the observed fairness issues in the system.

Fairness problems are often caused by:

- Data imbalance
- Indirect proxy features for protected attributes
- Measurement issues (biased measurements used as labels)

This step focuses on understanding the root cause of existing disparities before choosing a mitigation strategy. This ensures that interventions are effective, preventing misaligned fixes and thereby reducing both regulatory and operational risk.

What it Focuses On:

This Analysis evaluates fairness at the system level, not at the level of individual decisions.

It focuses on:

- Aggregate patterns across demographic groups
- Consistent disparities in outcomes, not isolated cases
- The relationships between data sources, features, and model outputs

The objective is to surface causal pathways that explain how disadvantage is introduced or amplified within the system.

Key Questions to Answer

Teams must answer:

- Which groups experience worse outcomes?
- Are some groups affected more when attributes combine (e.g., gender and ethnicity)?
- Which data or features may indirectly influence these outcomes?

These questions help identify the root cause of the issue. These questions guide teams toward understanding mechanisms, not just correlations.

Outputs

Completion of the Causal Fairness Analysis requires the following deliverables:

1. Causal Diagram

A simple visual representation showing relationships between:

- Protected attributes
- Input features
- Intermediate variable
- Model outcomes

The diagram should reflect the team's assumptions about how bias enters the system.

2. Root Cause Summary

A concise written explanation describing:

- Why does the observed disparity exist?
- Which factors contribute most strongly?
- Whether the issue originates from data, features, or measurement choices?

3. Affected Groups List

A clearly documented list of demographic and intersectional groups impacted by the disparity.

These outputs create a shared understanding across technical, product, and governance stakeholders

At the end of causal fairness analysis, teams must:

- Identify the stage(s) where bias enters or is amplified
- Map these stages to corresponding intervention strategies
- Document any technical or operational constraints
- Proceed to intervention selection with a causally justified plan

Cardinal Rule:

No intervention should proceed without this step.

Understanding the Causal Model

Types of Variables in the Causal Analysis

To reason clearly about fairness, the causal model distinguishes between different categories of variables:

Protected Attributes

These are characteristics that should not directly influence decisions and are protected under fairness, legal, or ethical standards.

Examples in this case study include:

- Gender
- Nationality or region of origin

Protected attributes are not necessarily used directly by the model, but they may still influence outcomes indirectly.

Proxy Variables

Proxy variables are features that are *not protected themselves* but correlate strongly with protected attributes and can transmit bias.

Examples include:

- Travel frequency (proxy for passport privilege)
- Language confidence scores (proxy for native language)
- OCR quality scores (proxy for script or document format)

Proxy variables are especially dangerous because they can encode protected information implicitly.

Intermediate Variables

These variables sit between inputs and outcomes and often represent system-generated signals.

Examples:

- OCR confidence
- NLP confidence
- Biometric match scores

Bias often enters through *measurement error* in these variables.

Outcome Variable

The final system decision, such as:

- Approval
- Manual review
- Rejection

The goal of the causal analysis is to understand how disadvantage propagates toward this outcome.

Types of Edges in the Causal Graph

Edges in the causal graph represent different kinds of influence:

- **Direct causal relationships**
 $A \rightarrow B$ indicates that changes in A directly affect B
(e.g., OCR quality $\rightarrow$ NLP confidence)
- **Assumed or contextual relationships**
These edges represent domain assumptions supported by prior evidence or expert knowledge
(e.g., nationality $\rightarrow$ document language)
- **Indirect causal pathways**
 $A \rightarrow C \rightarrow B$ shows how bias can flow through multiple steps even when no direct link exists
(e.g., nationality $\rightarrow$ OCR errors $\rightarrow$ lower confidence $\rightarrow$ rejection)

Not all edges represent certainty; some represent plausible mechanisms that must be tested or monitored.

Understanding Causal Graphs (Illustrative Example)

A clear understanding of what constitutes a comprehensive causal fairness analysis is essential before applying causal reasoning to the IVRAS context.

Illustrative Example – Generic Causal Pathway

Protected Attribute

Group Membership

↓ (assumed structural influence)

Proxy Variable

Neighborhood

↓ (direct causal effect)

Model Input

Income Stability Score

↓ (direct model computation)

Outcome

Loan Approval Decision

Explanation

This diagram illustrates how unfair outcomes can arise indirectly.

1. Group Membership (such as gender or ethnicity) may influence Neighborhood due to historical or structural inequality.
2. Neighborhood then influences Income Stability, which is directly used by the model. Income Stability affects the final decision.
3. There is no direct arrow from Group Membership to Loan Approval. However, an indirect pathway transmits disadvantage through the proxy variable.

This example demonstrates three important concepts:

Protected Attribute

A characteristic that should not directly determine outcomes.

Proxy Variable

A non-protected variable that is correlated with the protected attribute and may transmit bias.

Indirect Causal Pathway

A chain of influences through which disparities emerge even without explicit discrimination.

Causal analysis focuses on identifying and reasoning about these indirect pathways.

Application to IVRAS

The same causal reasoning was applied to the IVRAS system.

IVRAS Causal Example – Measurement Bias Pathway

Protected Attribute

Nationality

↓ (assumed structural influence)

Document Language

↓ (measurement effect)

OCR Confidence

↓ (direct model computation)

Risk Score

↓ (decision rule)
Rejection

Protected Attribute

Gender
↓ (interaction effect)
Risk Score

Explanation

In IVRAS, nationality influences document language.
Certain document languages are associated with lower OCR accuracy.
Lower OCR confidence reduces the system's overall risk score confidence.
Lower risk scores increase the likelihood of rejection.

Gender interacts at the risk score stage, which helps explain why disparities become more pronounced under intersectional analysis.

Importantly, there is no direct link from nationality to rejection.
The disparity emerges through measurement bias and model scoring.

This analysis supports the conclusion that the observed disparity is systemic and multi-stage, rather than the result of a single discriminatory rule.

The identified bias stage directly determines which intervention strategy is selected:

- Data-stage bias → Pre-processing interventions
- Model-stage bias → In-processing interventions
- Decision-stage bias → Post-processing interventions
- Multi-stage bias → Layered intervention approach

This ensures that all mitigation actions are causally justified rather than heuristic, preventing ineffective or misaligned interventions.

Causal Fairness Diagram (IVRAS Case Study)

Mermaid Diagram:

A[Nationality] --> B[Document Language]

B --> C[OCR Confidence]

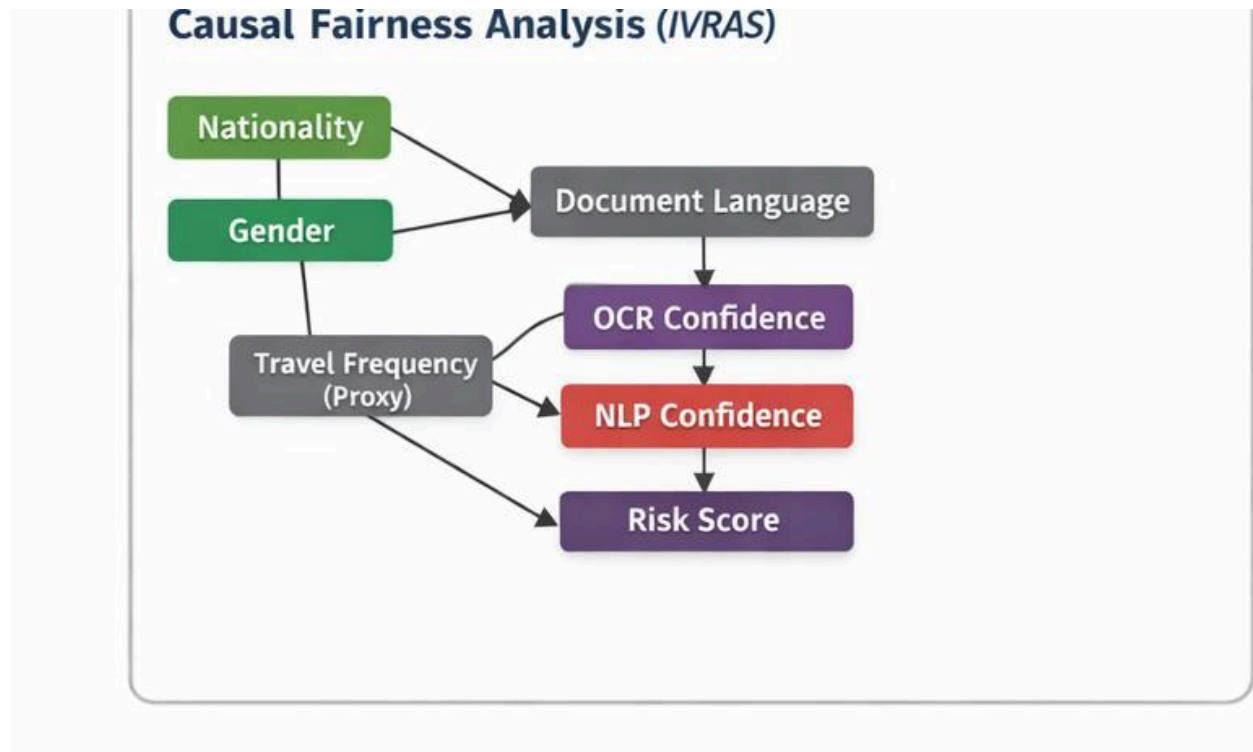
C --> D[NLP Confidence]

D --> E[Risk Score]

E --> F[Rejection]

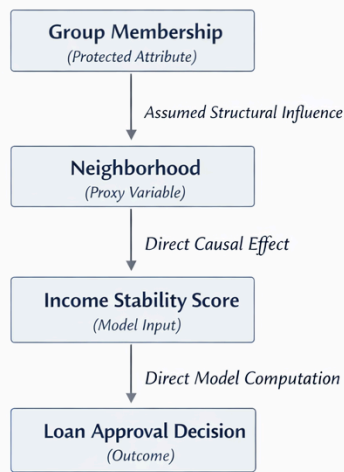
G[Gender] --> E

H[Travel Frequency (Proxy)] --> E



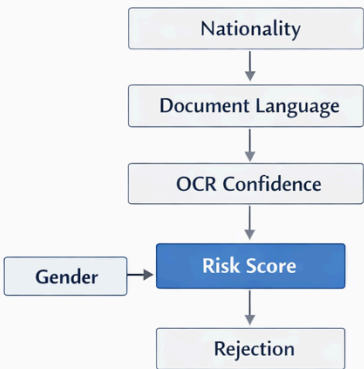
Below is the Diagram in visuals for better understanding

Illustrative Example



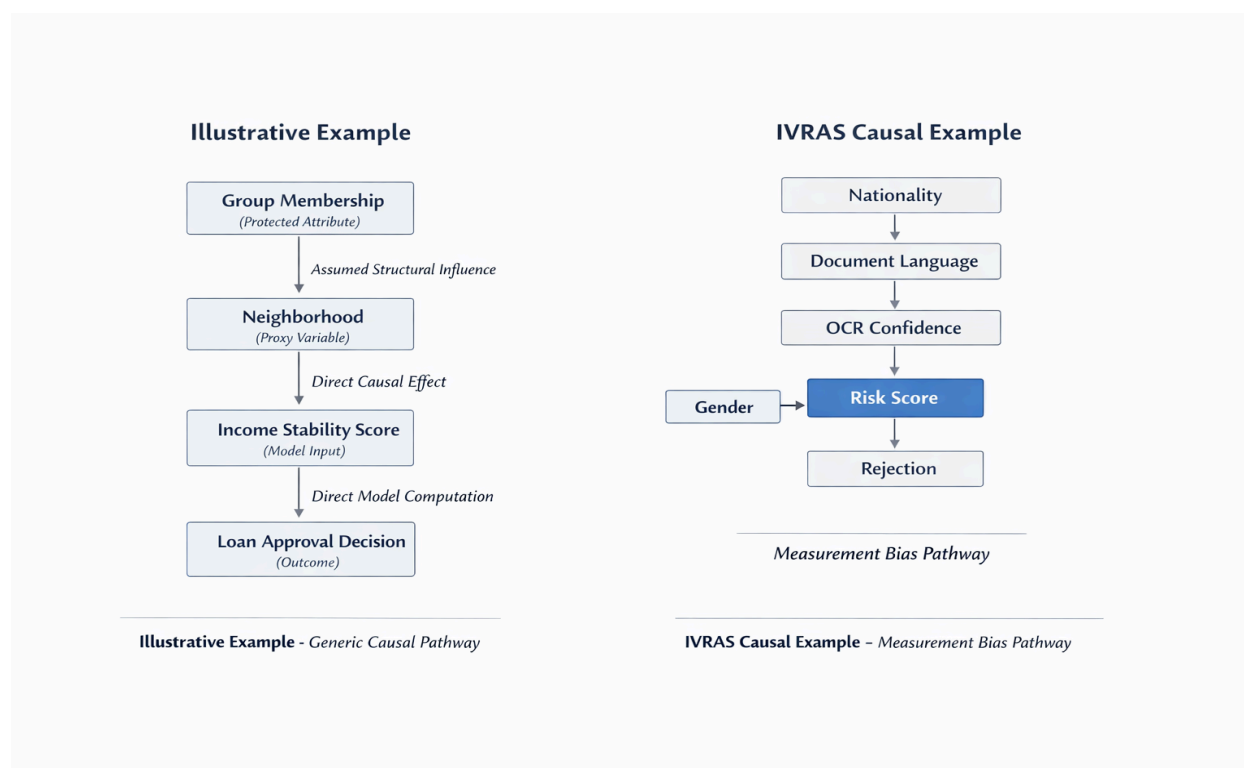
Illustrative Example - Generic Causal Pathway

IVRAS Causal Example



Measurement Bias Pathway

IVRAS Causal Example - Measurement Bias Pathway



Decision Impact Mapping

Stage	Bias Type	Impact
Data	Representation Bias	Underrepresentation of groups
Measurement	OCR/NLP bias	Lower confidence scores
Decision	Threshold bias	Unequal rejection rates

This mapping ensures that interventions are aligned with the stage where bias originates or is amplified.

Black-Box Implementation Example

Context

In some deployments, the internal model architecture, feature weights, or training pipeline are inaccessible.

This may occur when:

- The model is provided by a third-party vendor

- The model is proprietary
- Retraining is not permitted
- Deployment constraints prevent modification

In such cases, the system must be treated as a black box.

Only the following are observable:

- Input
- Outputs
- Group-level outcome disparities

Step 1 – Detect Disparity at Output Level

Observed:

- Higher rejection rates for Group A
- Higher false rejection rate compared to other groups

Since the internal mechanism is unknown, we begin with output-based analysis.

Step 2 – Perform Causal Hypothesis Mapping (External)

Even in a black-box setting, teams can construct a hypothetical causal graph based on domain knowledge.

Example:

Protected Attribute



Document Language



OCR Quality



Model Output (Risk Score)



Decision

Although internal weights are unknown, measurement disparities (e.g., OCR accuracy differences across scripts) can be externally tested.

Step 3 – Black-Box Testing Strategy

Because the model cannot be modified internally:

A. Counterfactual Testing

- Keep all features constant
- Change only protected attribute (if observable)
- Observe output changes

B. Sensitivity Analysis

- Slightly vary proxy features
- Measure impact on outcome

C. Threshold Sensitivity Testing

- Analyze output distribution near the decision boundary
- Identify concentrated harm

Step 4 – Intervention Options in Black-Box Setting

Since internal retraining is not possible:

Pre-Processing (Feasible)

- ☒ Improve input quality
- ☒ Normalize measurement differences
- ☒ Rebalance input distribution

In-Processing (Not Feasible)

- ☐ Cannot modify internal loss function
- ☐ Cannot alter feature weights

Documented as constrained.

Post-Processing (Primary Tool)

- ☒ Adjust decision threshold
- ☒ Introduce review band
- ☒ Apply reject-option fairness
- ☒ Route borderline cases to human review

Step 5 – Validation

Because model internals are unknown, validation must focus on:

- Error-rate parity
- Outcome disparity reduction
- Stability across time

If fairness improves without disproportionate performance loss, mitigation may proceed.

Governance Requirement

In black-box systems:

- In-processing may be unavailable
- Root-cause correction may be limited
- Post-processing must not mask systemic harm without documentation

All constraints must be formally recorded.

Output of Causal Analysis

The analysis must produce:

- Identified bias stage(s)
- Root cause summary
- Affected group mapping
- Causal pathway diagram

The identified bias stage feeds directly into the Risk-Based Intervention Selection Framework (see 03-risk-based-intervention-selection).